

INK & METAL



FTC 5773

PORTFOLIO

23 - 24

TEAM INTRODUCTION

We are **Team Ink and Metal 5773**, a 15-member team from American High School in Fremont, CA. Our philosophy maintains that every individual has the power to **shape the future** with their unique values and skills.



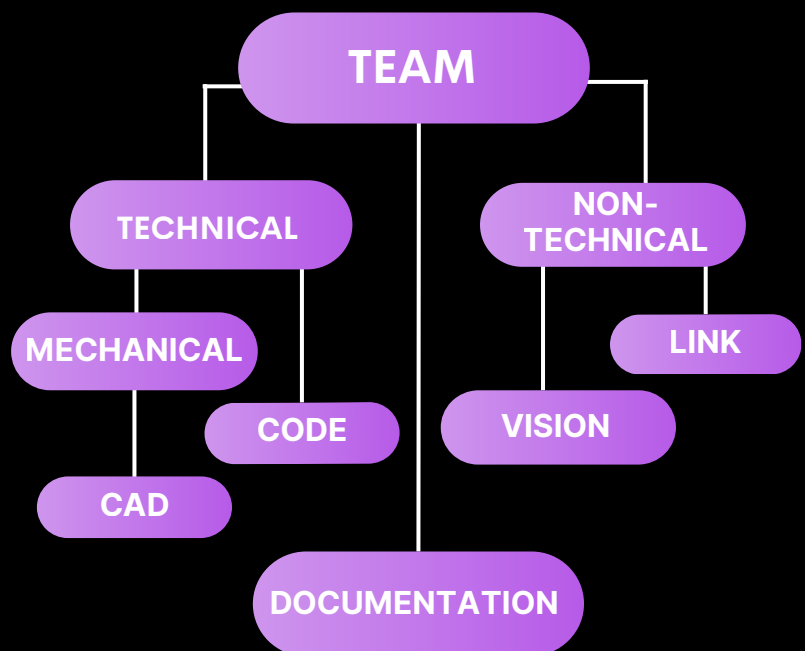
Over the past **13 years**, our team has consistently set ambitious goals to execute with our robot and community outreach. Each year we expand upon our knowledge to design **versatile mechanisms** for our robot, combining traditional concepts with innovative methods. Beyond our robotics endeavors, we aim to introduce FIRST to new people **around the world** to ignite their hidden skills and talents.

INNOVATE. IMPLEMENT. INSPIRE.

STRUCTURE

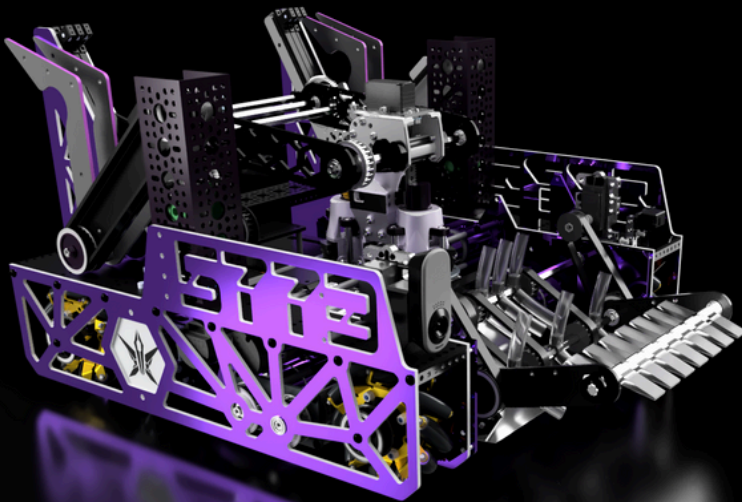
The team is divided into **six branches**, each with its respective Branch Captain. The Team Captain coordinates the workflow of the branches to meet the team's goals.

To ensure that the contributions are balanced, every member participates in a chosen technical and non-technical branch. We highly value **individual input**, and our branch system sets up the team for long-term **sustainability**.



OUR ROBOT

Our robot includes **five distinct subsystems**: chassis, intake, outtake, hang, and drone launcher. Combining the functionality of each individual component allows us to achieve all the tasks involved in the Center Stage challenge.



BENCHMARKING

Autonomous (**60** points)

- Score preloaded purple pixel on tape (**20** points)
- Score preloaded yellow pixel on the backdrop (**25** points)
- Score two white pixels (**10** points)
- Park robot (**5** points)

Tele-Op (**76** points)

- Score 12 pixels (**36** points)
- Create 2 mosaics (**20** points)
- Score 2 set bonus (**20** points)

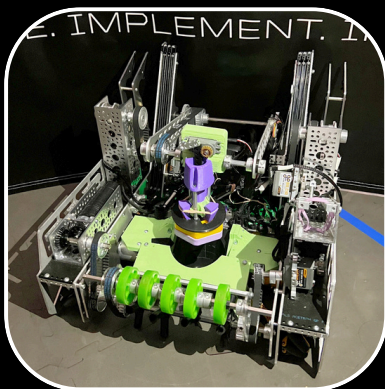
End Game (**50** points)

- Hang from truss (**20** points)
- Launch drone (**30** points)

Total Game Points: **186** points

ITERATIVE ROBOT EVOLUTION

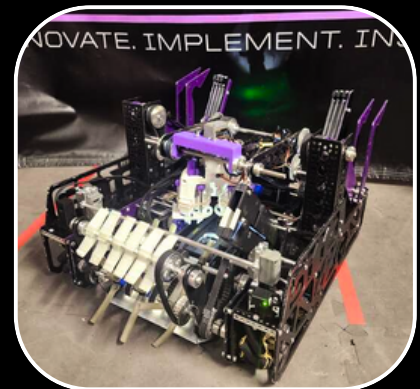
V1



CAD



V2



We iterate on the components of our robot meticulously. Every mechanism is brainstormed and designed, prototyped for a proof-of-concept with 3D printing, 3D modeled through CAD software, and finally built with aluminum CNC manufacturing.

GAME STRATEGY

This year's challenge requires our robot to **cycle across the field** and score swiftly. We designed quick and precise intake and outtake mechanisms to **best assist drivers**.

PLAN OF ACTION

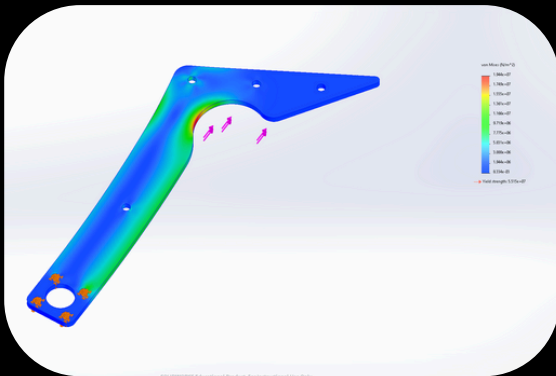
Autonomous: **Camera input** detects our custom element to score the purple and yellow **preloaded pixels**. We then collect and score 2 pixels from the pixel stack on the backdrop, and end with **parking**.

Tele-Op: We score pixels on the backdrop, aiming to **make mosaics** and **increase the height** of pixel structures for bonus points. Working with our alliance partner is crucial to reaching our scoring goal.

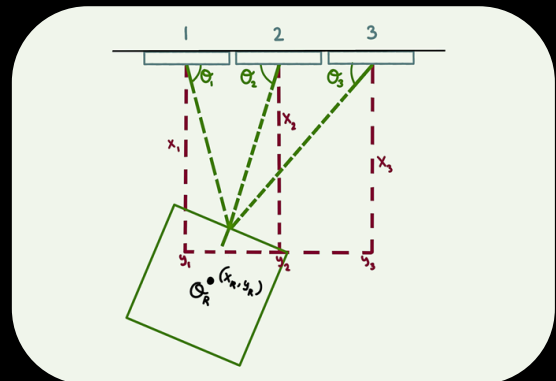
Endgame: We shoot our **drone** and passively **hang** on the truss.



CALCULATIONS



We used the **Solidworks Finite Element Analysis** to perform stress testing on crucial points of our robot. One such point was our hanging hook, which holds the mass of our robot during endgame. We replicated the hanging scenario to determine the need for reinforcements.



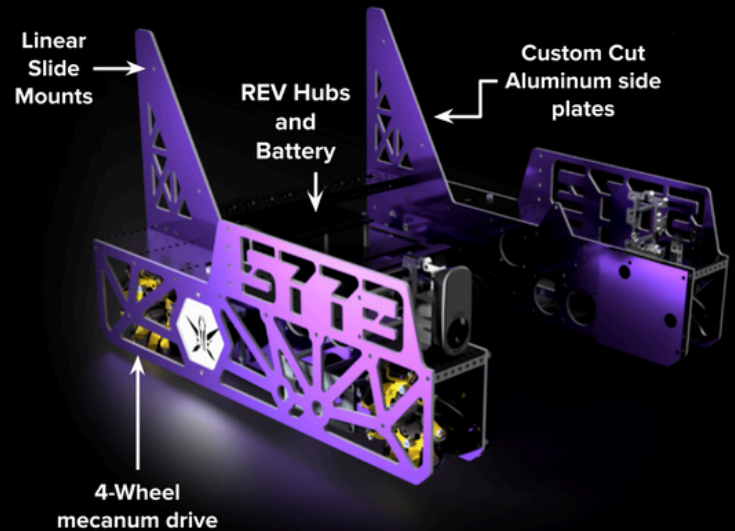
To fulfill our goal of minimizing driver effort, we **automated our depositing process** using the 3 April Tags present on the backdrop. Referencing the X, Y, and heading positioning of the tags, a push of a button aligns us with the backdrop perfectly.

CHASSIS

Our chassis can cycle between the corners of the field with quick, agile movement while minimizing drifting and tipping.

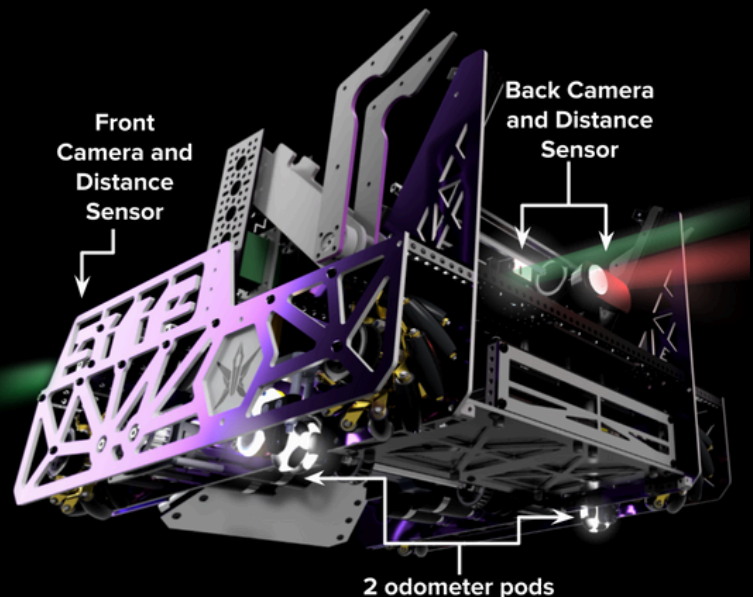
HARDWARE

The chassis utilizes a **four-wheel mecanum drivetrain**. Our robot's height of **11.5 inches** allows us to easily navigate under the gate. In previous years, we primarily used **aluminum** parts. This season, we switched multiple low-impact parts to **Delrin** to reduce weight and strain on our hanging mechanism. Our REV Hubs and battery are placed towards the bottom of the robot for a **low center of mass**. This increases stability without compromising on movement precision.



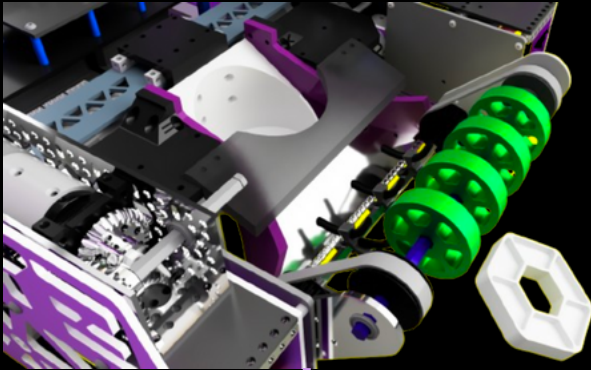
FIELD LOCALIZATION

To track and maintain the position of our robot during the autonomous period, we use **2 odometer pods** and the in-built Control Hub gyroscope to maintain our X, Y, and heading positions. Motion profiles accurately control our velocity through each movement while applying **PID-based position** control to account for any error accumulated during trajectories. Additionally, we apply principles from our open-sourced pathing algorithm, **BeepBeep**, which uses **Bezier curves** to make smooth, continuous paths around the field. This accurate movement allows us to travel with precision at high speed.



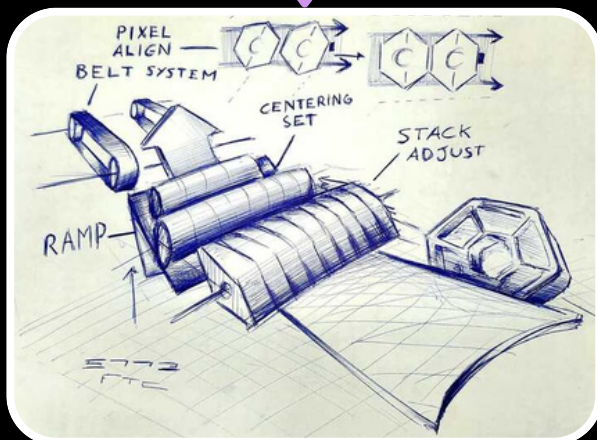
INTAKE

Our intake collects pixels from **stacks** or the **ground** into a conveyor belt within the robot to be transferred to the outtake system.



VERSION 1.0

The previous intake arm was **heavy** and **imbalanced**, leading to **poor control** when collecting from stacks. The stacking of pixels inside the robot required excessive precision, and **misaligned pixels** often caused **interferences** with the outtake.

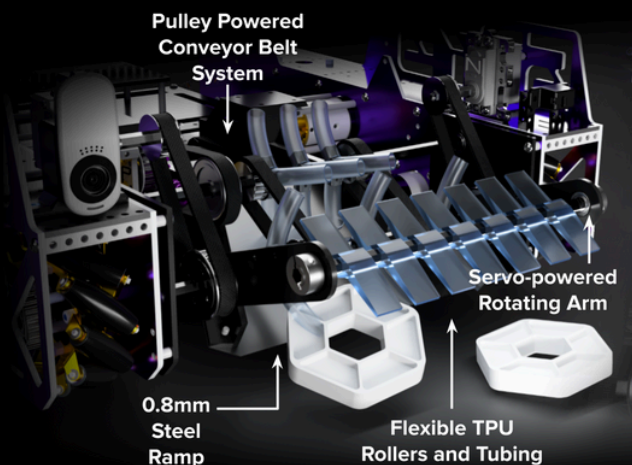


RE-ITERATE

- Accurately **control intake** from stacks
- Manage **pixel orientation** during the transfer to deposit
- **Reduce weight** of rotating arm
- Ability to **incorporate sensors** to automate intake
- Consistently **pull pixels** up the ramp

VERSION 2.0

Our current mechanism merges two subsystems for a streamlined intake. A dual servo-controlled arm with **flexible TPU flaps** matches the height of pixels in a stack. Using two rows of **surgical tubing**, pixels are pulled up a thin steel ramp into a **conveyor belt** system. The movement of belts provides a uniform force that pushes pixels into the correct orientation against a set of aligners. Two **color sensors** detect the presence of pixels once they are ready to be collected by the deposit.



OUTTAKE

An elastic powered end effector set on a servo-powered arm and diagonal linear stages serves as our outtake, combining precision, speed and versatility in its movements!

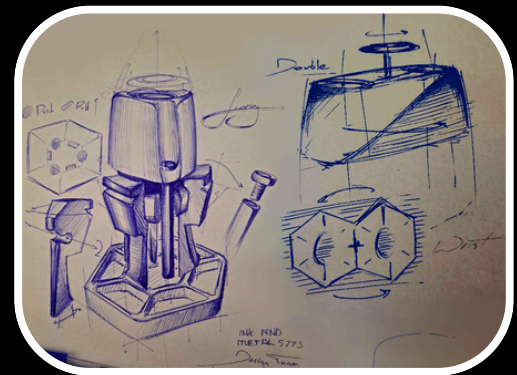
VERSION 1.0

In the first version, we designed a powered virtual 4-bar with an end effector that **carried two stacked pixels** and placed them precisely on the backboard. However, the end effector **lacked speed**, as pixels could only be dropped **one at a time** and required repositioning between each drop.



RE-ITERATE

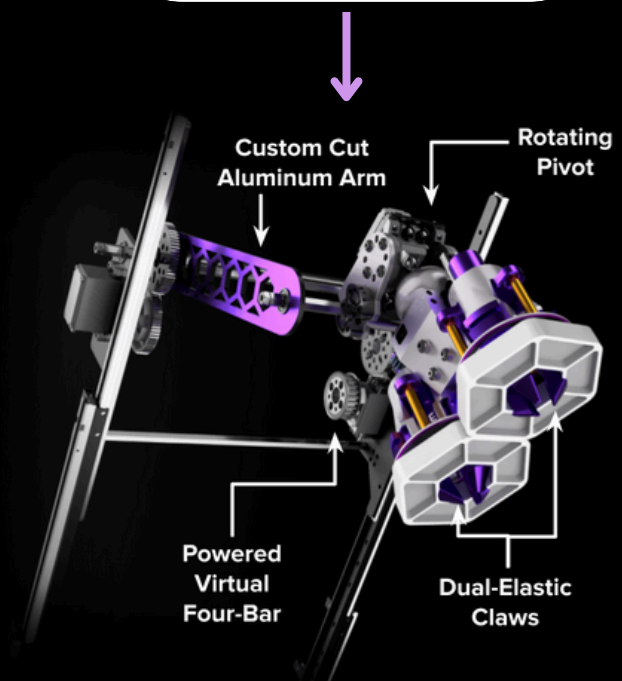
- Consider how the mechanism can balance **speed and precision**
- Consider **compatibility** with our new intake/transfer
- Minimize the number of design changes



VERSION 2.0

The deposit mechanism is mounted on a **virtual 4-bar** and extended by three stages of **linear slides**. It features **two claws** that grip pixels via **internal tension**. This is optimized for two functions:

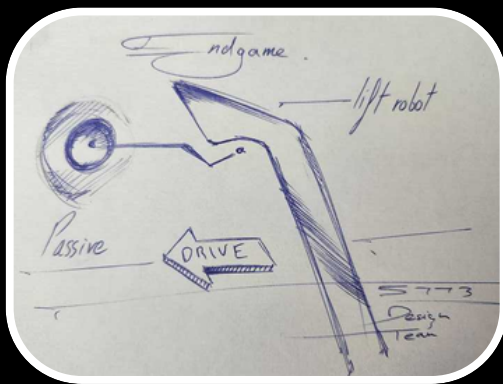
1. It can **drop two pixels side-by-side** at the same speed as a generic dump - **6 seconds**.
2. It can rotate into **any orientation**, allowing us to gain bonus points from scoring mosaics and set lines



HANGING

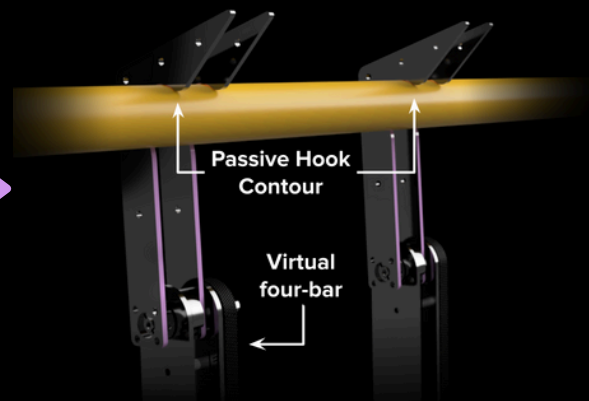
INITIAL IDEAS

- Positioned at the robot's **center of mass**
- Fit under the **12-inch height** constraint of the stage door
- Fold **compactly** into the frame of the robot
- Require **minimal movement** to extend



FINAL ITERATION

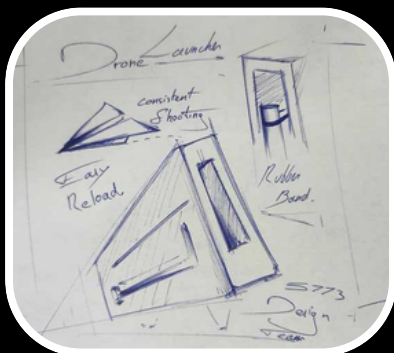
A **virtual four-bar** fulfills our height and center of mass constraints, keeping the robot level upon being elevated. The distinguishing feature of this hanging system is the **passive hook**; the chassis simply has to be driven into the truss to lift the robot.



DRONE LAUNCHER

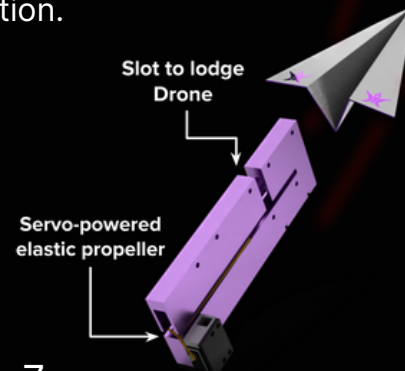
INITIAL IDEAS

- **Secure drone** against any external forces acting on the robot
- Launch angle must clear truss and land in Zone 1 for a bonus of **30 points** in the endgame
- Drone must be **aerodynamic** and resistant to gliding



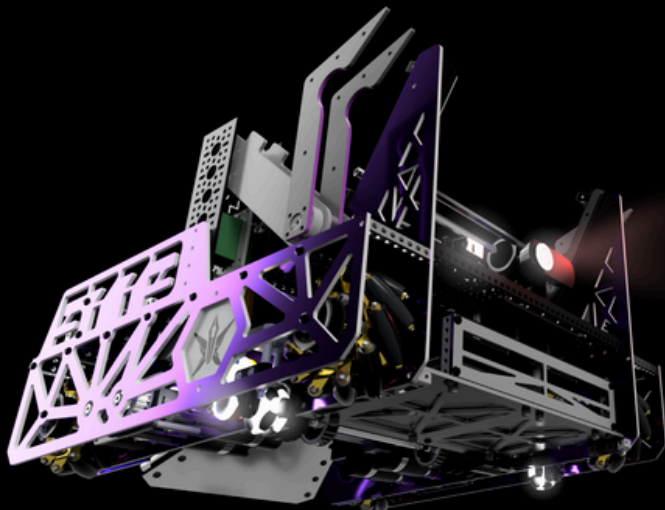
FINAL ITERATION

The drone is slotted into a **3D-printed** housing and held in place by an elastic. The launcher uses a servo to release the elastic and propel the drone over the truss. For the most optimized results, the launcher can be tested at **various elastic strengths** to collect data and finalize the position.



SOFTWARE ENHANCEMENTS

SENSOR ENHANCEMENTS

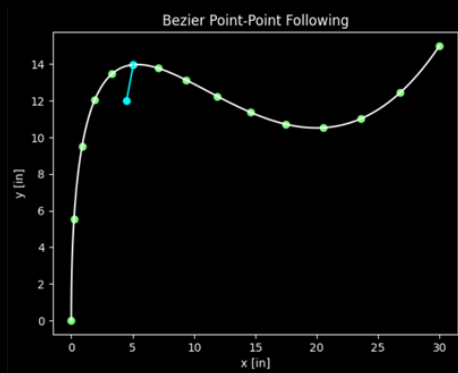


Autonomous: A **distance sensor** is used to space our robot with the stack to perfectly collect pixels **regardless of positioning error**.

Tele-Op: Align with the backdrop using our **camera** and **distance sensor** to **quickly** deposit pixels.

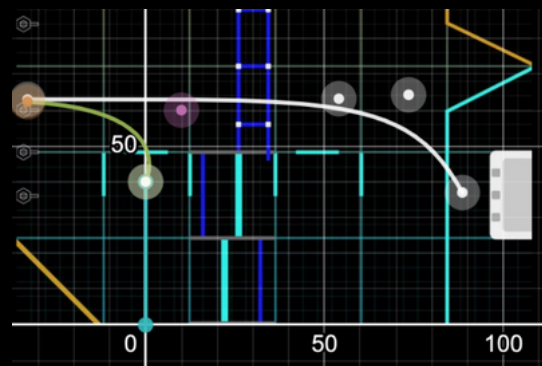
Endgame: Use **April Tags** on the backdrop to line up our airplane launcher, and finish the match by passively hooking onto the truss.

BEEPBEEP PATHING



The restrictive nature of current FTC autonomous movement libraries pushed us to develop **BeepBeep**: a custom path planning and motion profiling system that breaks down **Bezier curves** into a set of fluid motions for a robot to follow. Our library uses **position and velocity-based correction** to ensure the robot can tightly follow its path.

HOW IT WORKS



Using **4 points**, we produce a Bezier curve that forms our path. The length of the curve is calculated using calculus and is then broken down into equally spaced points. Using **position-based PID** and velocity-based feedforward control, our robot tightly follows the curve.

FIRST, CHANGE

FIRST IN GHANA

Our most impactful effort this season comes in our partnership with the Makerspace, an educational organization in **Ghana**. With no official region in their area, their students could not participate in FIRST. We raised **\$5,000** in parts to create **16 FIRST teams**, which we use in weekly meetings to guide them through concepts from mechanical design to financial sustainability. We are now establishing Ghana's **first-ever competitive region** in collaboration with FIRST delivery partner Coderina.



INTERNATIONAL FIRST DAY

We are working towards a federal resolution for National FIRST Robotics Day to celebrate FIRST's impact. To date, we've met with **9 US House Representatives** and **state governments** to pass the bill to establish robotics programs by distributing funding from the **Every Student Succeeds Act** for providing workspaces and resources. Now, our efforts have expanded to an International FIRST Day, involving **13 teams** from countries like Romania and China, where we are leading the effort to provide a total **\$27 million** for robotics programs **worldwide**.



STEM LOBBYING

Working towards **national change** showed us the need to address the outdated nature of current educational practices, motivating us to collaborate with the **US Department of Education** and the nationwide advocacy group **TIES** to modernize STEM education. We represented the perspective of FIRST robotics students, meeting with representatives to call for hands-on learning experiences for **42 million students** nationwide.



FIRST, TOGETHER



INKERNATIONAL

Intrigued by the diversity of teams' journeys from all corners of the world, we met with **121 international FIRST teams** to understand their experiences. Later, we partnered with them to produce **INKernational: A FIRST Documentary**, compiling the inspiring stories from every team we spoke to. The documentary has been published in **73 countries**, recognized by the Tri-City Voice Newspaper, and **18 FIRST regional directors**.

GLOBAL FIRST INCLUSIVITY CONFERENCE

Our desire to provide a platform for the diverse communities in FIRST and foster an open-minded community led to our global conference. We brought together **186 FIRST teams** to share their experiences in FIRST. Teams came from **all backgrounds**, including all-girls teams, disabled teams, LGBTQ+ teams, and racially diverse teams. Attendees were connected to a panel of **27 STEM professionals** who provided valuable advice and mentorship.



OUR FIRST JOURNEY

To share the memories, challenges, and insightful advice that our team's alumni have gained over the years, we published the **first-ever book about FIRST robotics: Our FIRST Journey**. Noting the success of our first edition, we published a second version with the updated stories of our recent alumni. Our book has been translated into **43 languages** and published across **143 locations** and 13 online platforms, including Barnes & Noble and Apple Books, garnering over **13,000 reads**.



FIRST, EDUCATION

ROBOTICS & CODING CLASSES

To ignite a passion for STEM in the next generation of our community, we conducted our **fourth annual Robotics and Coding Camp**, providing **256 local students** with skills in 3D Modeling, circuitry, Python, and Java. Students applied these skills to a prototyped robot. Pursuing global expansion, we established **six international robotics clubs** in Ghana, Dubai, and India, enabling **673 students** to experiment with differential drive robots with the help of virtual Webot simulations.



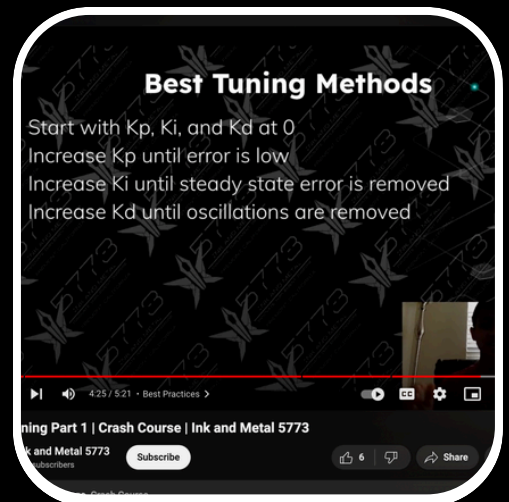
ROBOTICS ELECTIVE

Recognizing the lack of robotics programs available in schools, we implemented a **district-wide robotics elective** with the **Alameda County Board of Education**, providing students with FIRST-specific curricula including learning modules, team activities, and assessments. We open-sourced our engineering, programming, and AI curriculum via HeroX, a crowdsourcing organization, which has been shared with **2 million students** and **43 educational centers**.



CRASH COURSE VIDEOS

To foster FIRST education irrespective of background or location, we developed a **106-video FTC Crash Course** series, covering a diverse set of topics, including fundamental programming, 3D design, managing finances, and FIRST values. Through this effort, we created and are mentoring 11 FTC teams, 16 FLL teams, and 9 FLL chapters of the Silicon Valley Boys and Girls Club. Our series has accumulated over **80,000 views** and **790 subscribers**.



FIRST, DIVERSITY



FIRST FOR GIRLS IN STEM

Just 3 years ago, our team had only 2 girls, prompting our commitment to girls in STEM. Understanding the difficulty of having quality impact on distant regions, we harnessed collaboration in our **FIRST for Girls in STEM** initiative. We organized **18 teams** across the **US, China, India, and Spain** to come together to collectively bring STEM to young girls in unique educational opportunities. We have conducted summer camps, workshops, and online events for **207 female students**.



STEM-SIGN

Recognizing **a gap in STEM accessibility**, we expanded our mission to more underserved communities. We partnered with the **California Deaf School of Fremont**, not merely to host workshops, but to share STEM education to the hearing-impaired. Our team focused on teaching the **fundamentals of circuitry** in a uniquely adapted for the deaf students. Our team members even **learned ASL** to enrich the experience for everyone involved.



STANFORD SLAC RESEARCH

Wanting to bring direct and accessible research opportunities for girls interested in STEM, we worked with **Stanford University** to host four international workshops. Female researchers at the **Stanford National Accelerator Laboratory**, specializing in various STEM fields from **astrophysics to engineering**, presented to **371 students**. Students collaborated with the professors to develop **research projects** and present their findings.

CONNECTIONS

Connecting FIRST with the real world is essential to pushing ourselves past the limit each year. Our mentors, partners, and sponsors make our season possible with their valued support and guidance.

MENTORS

FRANCIS UNG

Francis Ung is an **alumni** of the team and UC Irvine graduate. As an alumnus, he has the expertise to guide us through FIRST-specific obstacles that we face. Additionally, his technical knowledge of **3D design** and **manufacturing** has proven invaluable to our robot design.



STEVE RODRIGGS

Steve Rodgers is a **Chief Engineer** at Lockheed Martin who helps us in **initial game discovery** and **robot brainstorming**. Using his industrial engineering skills, he provides new perspectives to approach each season. He teaches us how to apply the **design process** during prototyping, helping us improve each version of our robot.



DANNY DIAZ

Danny Diaz is a **FIRST Headquarters Senior Engineering Manager** who mentored us in **FTC logistical information**. He advises us on how to prepare for an FTC competition's judging presentation and portfolio. Additionally, he took part in our INKernational Documentary and Global FIRST Inclusivity Conference, extending his advice to other teams.



PROUD SPONSORS AND PARTNERS



Pacific Gas and
Electric Company



BUSINESS PLAN

Emphasizing sustainability, we focus on efficient financial management and a solid team structure to ensure we make responsible decisions that allow us to pursue our goals.

SWOT ANALYSIS



FINANCIAL SUSTAINABILITY

Our **school-affiliated** financial account limited opportunities to raise and access funds. Hence, we formed a partnership with Hack Bank, founding our own **fiscal non-profit** through which we manage expenses and fundraising. Our non-profit status verified our credibility within our community, resulting in multiple **new partnerships** with companies like Lockheed Martin and Apple, bringing us **\$5000** in sponsorships, grants, and donations. Managing our funds under Hack Bank taught members responsible and sustainable **money management skills** for the future.

FUTURE GOALS

- Utilize **custom path-generating software** to create the most optimal autonomous robot route using the Intel T265 Camera
- Use simulation software to replicate **real-life scenarios** based on CAD
- Start a **global WiSTEM organization** to come together to pursue projects and make an impact
- Expand our FIRST robotics elective to school districts **across the nation**

REFLECTION

Our team's dedication to **innovation and community impact** drives us to make **responsible decisions** and maintain a strong team structure. With proactive members, FIRST alumni, and local outreach connections, we aim to overcome challenges like team coordination. Our partnership with Hack Bank enhances our **financial stability**, empowering us to explore industrial solutions and expand robotics programs, reflecting our commitment to **positive change and resilience**.

INNOVATE. IMPLEMENT. INSPIRE.

These words define the motive and drive for our team. We pride ourselves in our positive attitude, regardless of whatever problems we encounter. As our strategy, skill, and team progress every season, these principles remain constant. Whether it is building a robot or teaching grade-school students about coding, we perpetuate this method:

Innovation, the action of using your knowledge and creativity to form effective solutions to problems.

Implementation, the ability to use the solutions within a given scenario effectively.

Inspiration, the spread of knowledge and gracious professionalism to the current and next generation of aspiring engineers.

We are Ink and Metal, and we are here to make an impact.

